

BUREAU OF INDIAN STANDARDS

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भारतीय मानक मसौदा
स्वास्थ्य सूचना विज्ञान — सार्वजनिक कुंजी ढांचा
भाग 4 स्वास्थ्य देखभाल दस्तावेजों के लिए डिजिटल हस्ताक्षर

Draft Indian Standard
Health Informatics — Public Key Infrastructure
Part 4 Digital Signatures for Healthcare Documents

[ICS 35.240.80]

Health Informatics Sectional Committee, MHD 17

Last date for comments:
18 September 2026

NATIONAL FOREWORD

(Adoption clause will be added later)

This standard supersedes IS/ISO 17090-4 : 2020 ‘Health informatics — Public key infrastructure: Part 4 Digital signatures for healthcare documents (*first revision*)’.

The text of ISO standard has been approved as suitable for publication as an Indian Standard without deviations. Certain conventions are however not identical to those used in Indian Standards. Attention is particularly drawn to the following:

- Wherever the words ‘International Standard’ appear referring to this standard, they should be read as ‘Indian Standard’; and
- Comma (,) has been used as a decimal marker while in Indian Standards, the current practice is to use a point (.) as the decimal marker.

In this adopted standard, reference appears to certain International Standards for which Indian Standards also exist. The corresponding Indian Standards which are to be substituted in their respective places are listed below along with their degree of equivalence for the editions indicated:

<i>International Standard</i>	<i>Corresponding Indian Standard</i>	<i>Degree of Equivalence</i>
ISO 17090-1 Health informatics — Public key infrastructure — Part 1: Overview of digital certificate services	IS/ISO 17090-1 : 2021 Health informatics — Public key infrastructure: Part 1 Overview of digital certificate services (<i>first revision</i>)	Identical

The Committee responsible for the preparation of this standard has reviewed the provisions of the following International Standards referred in this adopted standard and has decided that they are acceptable for use in conjunction with this standard:

International Standard	Title
ISO 14533-2 : 2021	Processes, data elements and documents in commerce, industry and administration — Long term signature — Part 2: Profiles for XML Advanced Electronic Signatures (XAdES)

Scope

This document supports interchangeability of digital signatures and the prevention of incorrect or illegal digital signatures by providing minimum requirements and formats for generating and verifying digital signatures and related certificates.

This document describes the common technical, operational, and policy requirements to enable digital certificates to be used in protecting the exchange of healthcare information within a single domain, between domains, and across jurisdictional boundaries. The purpose of this document is to create a platform for global interoperability. It specifically supports digital certificate enabled communication across borders but can also provide guidance for the national or regional deployment of digital certificates in healthcare.

This document defines the provable compliance with a public key infrastructure (PKI) policy necessary in the domain of healthcare. It specifies a method of adopting long-term signature formats to ensure integrity and non-repudiation in long-term electronic preservation of healthcare information.

This document provides healthcare-specific PKI (HPKI) profiles of digital signature based on the ISO/ETSI standard profiles specified in CAdES (CMS Advanced Electronic Signature), XAdES (XML Advanced Electronic Signature), PAdES (PDF Advanced Electronic Signature) and the ETSI standard specified in JAdES (JSON Advanced Electronic Signature).

The technical content of the document has not been enclosed as it is identical with the corresponding ISO Technical Specification. For details, please refer to ISO 17090-4 : 2026 or kindly contact:

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